Computational Literature Using Python

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Northeastern University
NULab for Digital Humanities and
Computational Social Science

Feel free to ask questions at any point during the presentation!

Agenda

- Different Methods for Computational Storytelling
- Introduction to Python and Google Colab
- Hands on: Using Computation in Writing
- Further Resources and Notes on Artificial Intelligence

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Teaching Materials

Slides, Python notebooks, and class activities can be found in the shared Google Drive folder titled SP26-PythonLit-StudentAccess at:

bit.ly/sp26-young-CRWT1801-pythonlit

To run the Python notebooks in Google Colab you will need to sign into Google Drive.

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Different Methods for Computational Storytelling

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Algorithms

- "[a] set of instructions that is designed to accomplish a task" (National Library of Medicine).
- A recipe for baking cookies, code to tell whether a picture is of a cat or a dog, and code to write a poem about a cat are examples of algorithms

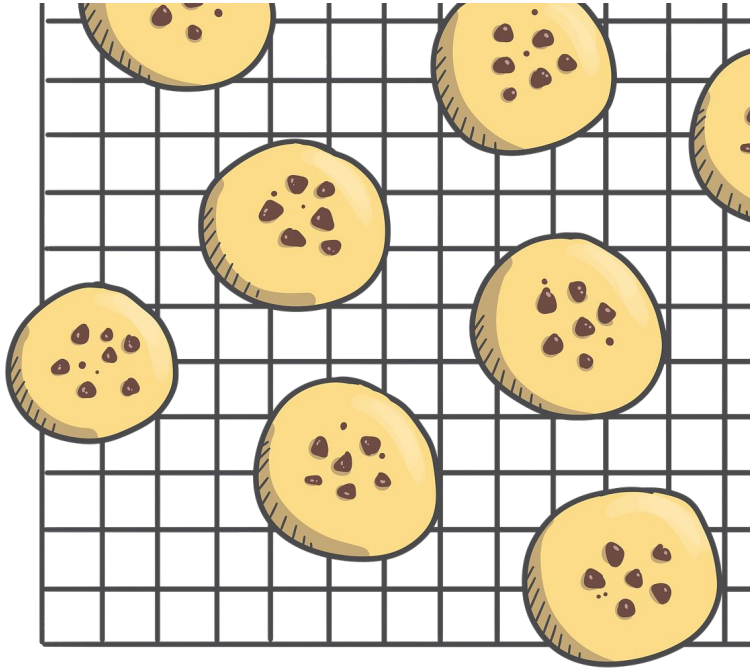

Image by wixin_56k, Pixabay

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Example: AI Rhyming Clock

- Claude's AI Rhyming Clock
  - Uses generative AI to write a poem integrating the current time
- Try going to the link above and clicking customize to adjust the prompt generating the poems

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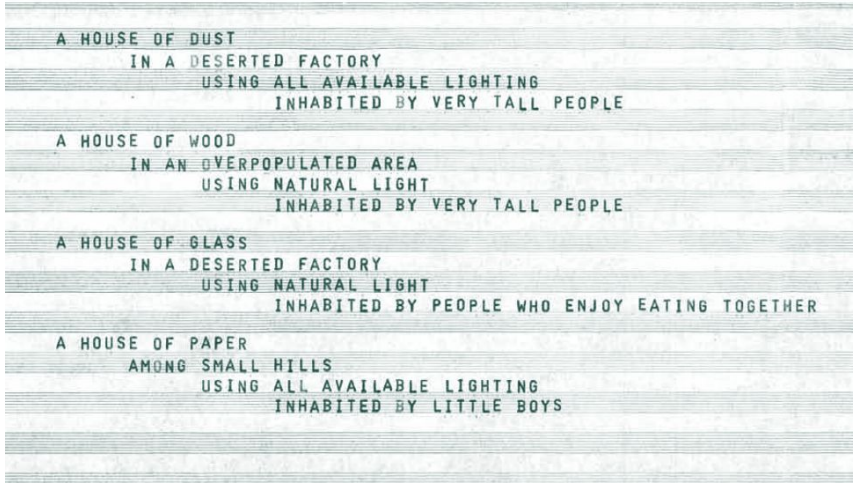
Example: “The House of Dust”

- Poem by Alison Knowles and James Tenney (1967)
- Code reimplemented in Python by Nick Montfort and updated as teaching example: [House of Dust Example]



A HOUSE OF DUST
IN A DESERTED FACTORY
USING ALL AVAILABLE LIGHTING
INHABITED BY VERY TALL PEOPLE

A HOUSE OF WOOD
IN AN OVERPOPULATED AREA
USING NATURAL LIGHT
INHABITED BY VERY TALL PEOPLE

A HOUSE OF GLASS
IN A DESERTED FACTORY
USING NATURAL LIGHT
INHABITED BY PEOPLE WHO ENJOY EATING TOGETHER

A HOUSE OF PAPER
AMONG SMALL HILLS
USING ALL AVAILABLE LIGHTING
INHABITED BY LITTLE BOYS

Printout of “The House of Dust,” Gebr König Verlag, Cologne, 1967

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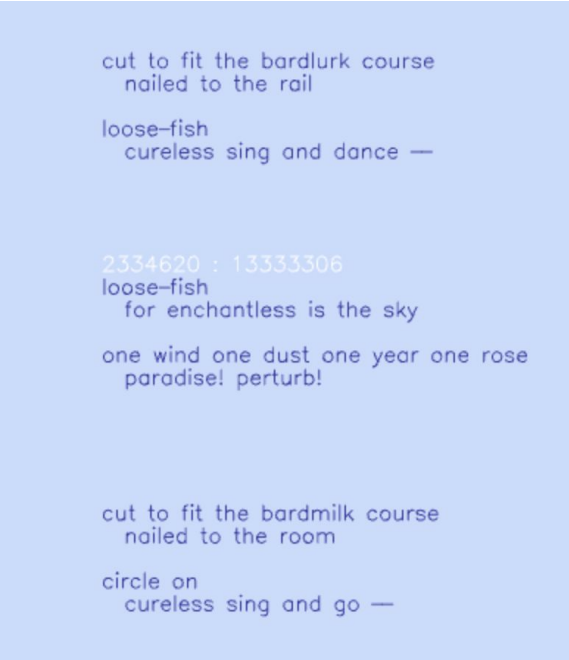
Example: “Sea and Spar Between”

Textual remix/deformance

[How to Read Sea and Spar Between]():

“The words in Sea and Spar Between come from Emily Dickinson’s poems and Herman Melville’s Moby Dick.”

-by Nick Montfort and Stephanie Strickland



Sea and Spar Between

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Generative AI & Computational Writing

Compare the 1) poetry generated using AI in Claude's AI Rhyming Clock to the 2) computationally determined poems like "House of Dust" and "Sea and Spar Between"

- How does computational poetry differ from poetry written using generative AI?

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Introduction to Python and Google Colab

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Python Summary

The Python code in this workshop covers these topics:

Data

- Variables
- Data types
  - Strings
  - Lists
- Selecting data from lists and dictionaries

Functions

- Built-in functions
- Importing modules
  - Importing functions

Conditionals

- If/else statements

Loops

- While loop

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Python Google Colab Notebooks

Intro To Python Poetry: This notebook introduces the fundamentals of Python and provides example code for creating computational poetry. The notebook can be accessed by:

- Signing in to your Google account
- Clicking the above link or navigating to the notebook in the shared Google Drive folder (notebook file name is SP26_Young_Colab&IntroToPythonLit.ipynb)
- Copying the notebook to your Google Drive by selecting 'Save a copy in Drive' under the 'File' menu in the upper left corner of the notebook

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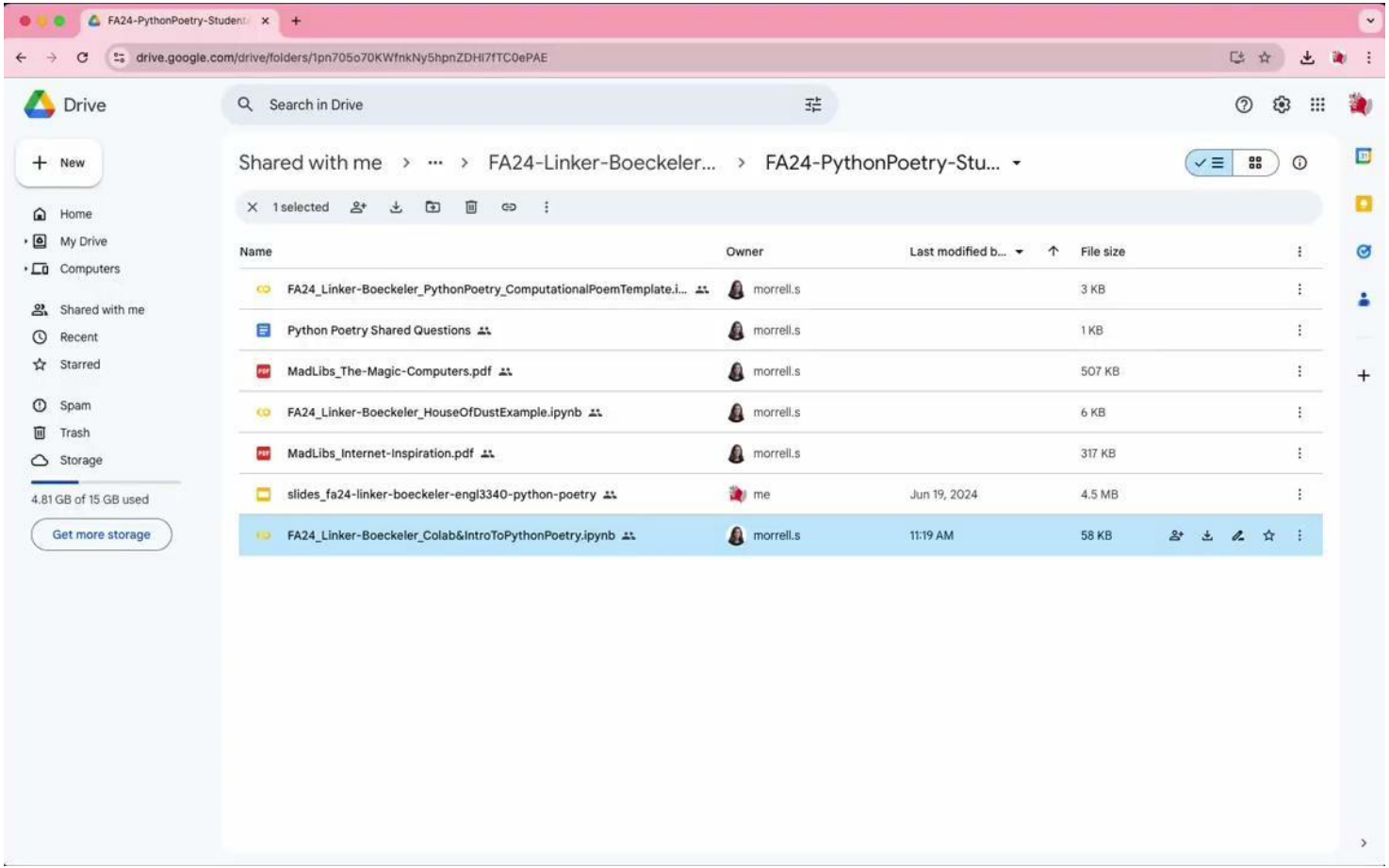
Install Google Colab



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Variables

Variables are created when you assign a value to them. The value can be **numbers**, **strings**, or another data type.

E.g. `x = 5, y = "Hello, world!"`

Strings

Strings in python are sequence of characters surrounded by either single quotation marks, or double quotation marks.

`'hello'` is the same as `"hello"`.

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Lists

Lists are used to store multiple items in a single variable. Items can be duplicated.

E.g. `fruit_list = ["apple", "banana", "cherry", "apple"]`

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Selecting Data from Lists

- List items are ordered, changeable, and allow duplicate values.
- List items are indexed, the first item has index `[0]`, the second item has index `[1]` etc.

Code

```
thislist = ["apple", "ball", "cat"]
print(thislist[0])
```

Result

```
apple
```


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Functions

- A function is a block of code which only runs when it is called.
- You can pass data into a function.
- A function can return data as a result.
- For example, you can display a string with the **`print()`** function:
- E.g. **`print("Hello")`**

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Python modules

- A module contains functions that someone else has written that you can use.
- When we want to use a module, we import it. For example, we are going to use the **`random`** module.
  - Code: **`import random`**
- We want to use a particular function, **`choice()`**, which gives us a random choice from a list or other data structure. We can import this function from the **`random`** module.
  - Code: **`from random import choice`**

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Conditionals

- Conditionals can be used to execute code if a condition is true
- To check if a condition is true use **==**
- Use **if**, **elif**, and **else** to designate the code you want to execute

Code

```
color = "pink"
if color == "pink":
   print("the the color is pink")
elif color == "blue"
   print("the the color is blue")
else
   print("the color is not pink or blue")
```

Result

```
the color is pink
```


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Loops

- Loops execute code repeatedly
- A **`while`** loop executes code while a condition is true
- A while loop needs to reach a point where the condition is not longer true, or it will continue running

Code

```
while condition == True:
    print("the condition is true")
    condition = False
```

Result

```
the condition is true
```


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Important notes about coding

- Capitalization matters
- Spaces and indentations matter
- Spelling matters
- Missing one quote mark/parentheses, mispairing quote marks matter
- If your code doesn't run, check the above first.
- It is okay to make mistakes

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Hands-On: Using Computation in Prose or Poetry

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Student Examples: Computational Prose

- Original writing: Funny Letter Example
- Remixing literature: Brontean Letter Example

Reflection Questions: How do you think computational remixing changes across genres? Which of these examples do you think worked better and why?

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You Try! (Prose)

1. Make a **copy** of either the funny letter or Brontean letter notebook
2. Make a change to the inputs in each of the text lists
3. Make at least **one** change to the story **structure**, either by adding a new list for an extra line, swapping the order of story elements, etc., and see what comes out!
    a. You can either use the text provided in the examples throughout this module or input your own generated in class

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You Try! (Poetry)

1. Make a copy of either the House of Dust Introductory notebook or the House of Dust Advanced notebook.
2. Make a change to the inputs in each of the text lists.
3. Make at least **one** change to the poem **structure**, either by adding a new list for an extra line, swapping the order of poem elements, etc., and see what comes out!
4. Try adding a pause or user input to the poem.

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Post-exploration group reflection

- Do you have any reflections on using Python for constructing poetry/prose?
- How does the writing process differ from traditional storytelling?
- How might the code impact the readers' perception of the poem?
- If you could design any creative writing tool using computational methods, what type of tool would want to design?
- How might you use this in the future?

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Further Resources and Notes on Artificial Intelligence

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Discussion: Generative AI in Computational Writing

- You can prompt GenAI to generate sentences, paragraphs, or complete code for your computational writing
- It is important to verify output and understand generated code
- How might using GenAI shape your computational writing?
  - Is it still **your** writing?
- What are the risks associated with using GenAI?
- If GenAI can write code, why might we still want to learn to code ourselves?

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Generative AI Example

- Example prompt: “Fill in the options with fun poem lines. Here is the code: [your code]”
- Result:

Copy Edit

```
poetry_lines = {
    'space_adventures': {
        'line1': [
            "Rocket ships zoom past stars with a grin,",
            "Through Saturn's rings we bravely spin,"
        ],
        'line2': [
            "Aliens wave as we fly by their moon,",
            "We dance with comets to a Martian tune."
        ]
        # To add more lines, you could add 'line3': ["option1", "option2"]
    },
    'lazy_sundays': {
        'line1': [
            "Sunlight spills across my cozy bed,",
            "Pancakes stack higher than my sleepy head,"
        ],
        'line2': [
            "I drift in dreams while the coffee brews,",
            "Wrapped in a blanket, I snooze and snooze."
        ]
        # Add 'line3', 'line4', etc., for a longer poem
    }
    # To add more topics, just add another key like 'underwater_worlds': {...}
}
```


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AI Ethics Resources

- The Institute for Experiential AI at Northeastern
- A Human Rights-Based Approach to Responsible AI by Vinodkumar Prabhakaran, Margaret Mitchell, Timnit Gebru, Iason Gabriel
- Ethics of AI, United Nations Educational, Scientific and Cultural Organization (UNESCO)
- The <A+> Alliance for Inclusive Algorithms
- Algorithmic Justice League

Image: Changing the Landscape of AI Governance, UNESCO

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Further Exploration:

How might you further randomize stories in Python? Here are some more resources to help you get started:

- 10 Fun Python Scripts That Generate Random Stories by *Medium*
- Code Club: Story Time by *Raspberry Pi Foundation*
- How to Build a Random Story Generator Using Python by *Geeks for Geeks*

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Handouts

- Writing Functions in Python
- Troubleshooting Python

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Thank you!

—Developed by Sara Morrell, Dipa Desai, Kasya O'Connor Grant, Avery Blankenship, Claire Lavarreda, Emily Sullivan, and Zhen Guo

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- If you have any questions, contact us at: nulab.info@gmail.com
- We'd love your feedback! Please fill out a short survey here: https://bit.ly/diti-feedback

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